

Section 5.5 — Regime-Dependent Analysis

The Plumbing of Liquidity: How Funding Conditions Transmit to Credit Spreads

Jordi Camps Triay

May 13, 2026

Contents

1	Introduction	2
2	Regime Classification	2
2.1	Threshold Definition	2
2.2	Stress Episode Identification	3
2.3	Stress Episode Timeline	3
3	Conditional Summary Statistics	4
4	Regime-Dependent Granger Causality	4
4.1	Stress-Period Results	4
4.2	Calm-Period Results	5
4.3	The Regime Contrast	7
5	Rolling-Window Correlation	7
6	Scatter Plots by Regime	7
7	Interpretation	8
7.1	The Channel Rotation Mechanism	8
7.2	Credit Spread Volatility Amplification	11
7.3	Implications for the Unconditional VAR	11
8	Robustness: Threshold Sensitivity	11
9	References	12

1 Introduction

The unconditional VAR analysis (Section 5.4) documented that funding variables explain only 2–3% of credit spread forecast error variance in the post-2018 sample, rising to 3.8% when the TED spread replaces SOFR–EFFR. However, the distributional analysis in Section 5.2 showed that the SOFR–EFFR spread is compressed at or near zero on most trading days (skewness = 22.5), with meaningful departures occurring only during episodic funding stress events. This suggests that the funding-to-credit transmission channel may be fundamentally nonlinear: its behaviour during genuine dislocations may differ qualitatively from its behaviour during normal money market operations.

This section tests that hypothesis by splitting the post-2018 sample into stress and calm regimes based on the absolute SOFR–EFFR deviation and examining whether the Granger causality results documented in Section 5.3 are regime-dependent. The results reveal a more nuanced picture than simple “stress-on / calm-off”: multiple funding channels operate during normal conditions, while only the collateral-quality channel survives during genuine dislocations.

2 Regime Classification

2.1 Threshold Definition

We classify each trading day as belonging to the **stress** or **calm** regime based on whether $|\text{SOFR} - \text{EFFR}|$ exceeds a threshold of 5 basis points (0.05 percentage points). This threshold is designed to capture genuine funding market dislocations rather than routine rate fluctuations. Under the Federal Reserve’s ample-reserves framework, the SOFR–EFFR spread is anchored within a narrow corridor determined by the interest on reserve balances (IORB) and the overnight reverse repurchase (ON RRP) rate. Small deviations within 5 bp reflect normal day-to-day variation in repo market supply and demand; departures beyond 5 bp signal that collateral scarcity, reserve drainage, or some other structural friction is materially affecting funding prices.

Table 1: Regime Classification Sensitivity to Threshold

Threshold ($ \text{SOFR} - \text{EFFR} >$)	Stress Days	Share
1 bp	1458	72.2%
2 bp	1212	60.1%
5 bp (primary)	313	15.5%

Under the 5 bp threshold, 313 days (15.5%) are classified as stress and 1,705 days (84.5%) as calm. This asymmetric split is economically appropriate: genuine funding dislocations should be rare events in a well-functioning ample-reserves system, not the norm. By contrast, a 2 bp threshold classifies 60% of the sample as stress—an implausibly high proportion that conflates routine rate fluctuations with genuine dislocations. The 5 bp threshold isolates episodes such as the September 2019 repo crisis ($\text{SOFR} - \text{EFFR} = 295$ bp), the March 2020 COVID dash-for-cash, quarter-end pressures, and the 2022 quantitative tightening period.

2.2 Stress Episode Identification

Table 2: Selected Major Stress Episodes ($|\text{SOFR}-\text{EFFR}| > 5 \text{ bp}$, $\geq 5 \text{ days}$)

Start	Event	Days	Max $ \text{SOFR}-\text{EFFR} $	Cum. ΔBaa	Cum. ΔAaa
2018-12-28	Quarter-end	6	0.75	+0.03	-0.02
2019-06-26	Quarter-end	6	0.17	-0.11	-0.13
2019-09-12	Sep 2019 repo crisis	5	2.95	-0.04	-0.04
2020-03-11	COVID dash-for-cash	13	0.29	+1.00	+0.12
2021-03-11	Post-SLR expiry	34	0.06	-0.14	-0.16
2022-06-07	QT/rate hiking	18	0.14	+0.34	+0.28
2022-09-07	QT/rate hiking	19	0.12	+0.02	-0.05
2025-09-25	Quarter-end	39	0.36	+0.05	+0.07

Cumulative changes in percentage points. Max SOFR-EFFR in percentage points.

The stress episode table reveals heterogeneity in outcomes. The March 2020 COVID episode produced a massive +100 bp cumulative Baa widening but only +12 bp for Aaa—consistent with credit-risk differentiation during a systemic shock. The September 2019 repo crisis, despite producing the largest SOFR-EFFR spike (295 bp), generated only modest credit spread movements (−4 bp for both Aaa and Baa), suggesting that repo dislocations can be contained within the money market without fully transmitting to credit. The 2022 QT episodes show mixed-sign credit responses, reflecting the competing forces of monetary tightening (widening) and economic resilience (tightening).

2.3 Stress Episode Timeline

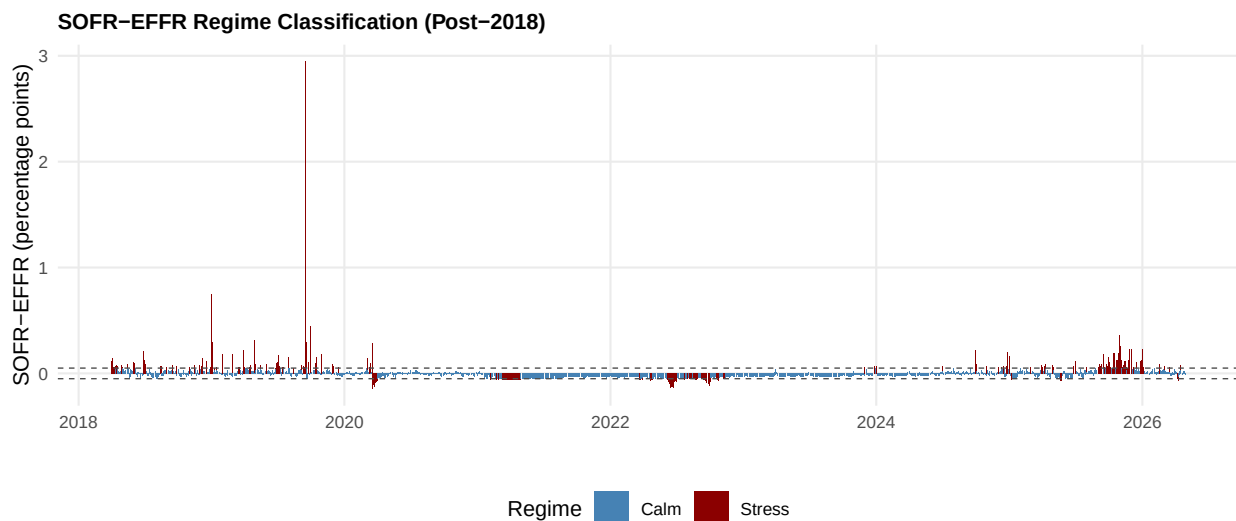


Figure 1: SOFR-EFFR with regime classification (Post-2018). Red bars indicate stress days ($|\text{SOFR}-\text{EFFR}| > 5 \text{ bp}$). The September 2019 spike (2.95 pp) and the March 2020 COVID episode are the most prominent stress events.

3 Conditional Summary Statistics

Table 3: Summary Statistics by Regime (Post-2018, 5 bp Threshold)

Variable	Calm (1,705 days)		Stress (313 days)		Var. Ratio
	Mean	SD	Mean	SD	
SOFR–EFFR	−0.011	0.024	0.047	0.199	—
ΔBaa	−0.0009	0.030	0.0050	0.044	2.14***
ΔAaa	−0.0002	0.034	0.0018	0.061	3.20***
$\Delta(Baa-Aaa)$	−0.0007	0.023	0.0031	0.040	2.99***
$\Delta Reserves$	1,190	35,564	−3,857	53,752	2.29***

Variance ratio = $\text{Var}(\text{Stress})/\text{Var}(\text{Calm})$. *** denotes F-test rejection at 1 percent level. Reserves in millions USD.

The regime-conditional statistics reveal sharp contrasts. SOFR–EFFR volatility is 8 times higher in the stress regime ($SD = 0.199$ vs. 0.024), confirming that the 5 bp threshold effectively separates tranquil from volatile funding conditions. More importantly, credit spread volatility is dramatically amplified during stress:

- **ΔAaa variance is 3.2 times higher** during stress (F -test $p < 0.001$), the largest amplification factor among the credit variables.
- **ΔBaa variance is 2.1 times higher** during stress ($p < 0.001$).
- **$\Delta(Baa-Aaa)$ variance is 3.0 times higher** during stress ($p < 0.001$), indicating that the quality differential itself becomes much more volatile during funding dislocations.

These variance ratios are substantially larger than those obtained with a 2 bp threshold, confirming that the 5 bp cutoff more cleanly separates high-volatility stress episodes from normal operations.

The mean ΔBaa during stress is $+0.50$ bp/day, significantly positive (t -test $p = 0.024$), while Aaa means are not significantly different across regimes ($p = 0.56$). This asymmetry—Baa spreads widen on average during stress while Aaa does not—is consistent with credit-risk differentiation: during funding dislocations, lower-grade credit absorbs more of the repricing.

4 Regime-Dependent Granger Causality

This is the core analysis: we replicate the Granger causality tests from Section 5.3 separately for stress and calm periods.

4.1 Stress-Period Results

Table 4: Granger Causality Tests — Stress Regime ($|\text{SOFR-EFFR}| > 5$ bp)

Direction	Lag	ΔN	ΔF -stat	Δp -value
SOFR–EFFR $\rightarrow \Delta Baa$	1	309	0.361	0.5483

SOFR–EFFR $\rightarrow$ Δ Baa	5	309	0.530	0.7537	NA
SOFR–EFFR $\rightarrow$ Δ Aaa	1	309	0.011	0.9167	NA
SOFR–EFFR $\rightarrow$ Δ Aaa	5	309	0.635	0.6730	NA
SOFR–EFFR $\rightarrow$ Δ (Baa–Aaa)	1	309	1.934	0.1654	NA
SOFR–EFFR $\rightarrow$ Δ (Baa–Aaa)	5	309	1.188	0.3151	NA
Δ Reserves $\rightarrow$ Δ Baa	1	309	0.042	0.8373	NA
Δ Reserves $\rightarrow$ Δ Baa	5	309	0.133	0.9847	NA
Δ Reserves $\rightarrow$ Δ Aaa	1	309	1.917	0.1672	NA
Δ Reserves $\rightarrow$ Δ Aaa	5	309	1.376	0.2334	NA
Δ Reserves $\rightarrow$ Δ (Baa–Aaa)	1	309	6.571	0.0108	**
Δ Reserves $\rightarrow$ Δ (Baa–Aaa)	5	309	2.706	0.0208	**
Δ TGA $\rightarrow$ Δ Baa	1	309	0.000	0.9979	NA
Δ TGA $\rightarrow$ Δ Baa	5	309	0.093	0.9932	NA
Δ TGA $\rightarrow$ Δ Aaa	1	309	0.122	0.7271	NA
Δ TGA $\rightarrow$ Δ Aaa	5	309	0.285	0.9213	NA
Δ TGA $\rightarrow$ Δ (Baa–Aaa)	1	309	0.000	0.9931	NA
Δ TGA $\rightarrow$ Δ (Baa–Aaa)	5	309	0.520	0.7610	NA

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Tests restricted to stress-regime days only ($N = 309$ after differencing).

During stress periods (309 observations after differencing), most Granger tests are insignificant. The one exception is the collateral-quality channel:

- **Δ Reserves $\rightarrow$ Δ (Baa–Aaa)**: Significant at both lags ($F = 6.57$, $p = 0.011$ at lag 1; $F = 2.71$, $p = 0.021$ at lag 5). During genuine funding dislocations, reserve changes predict the credit quality differential—consistent with the collateral hierarchy mechanism where Aaa bonds, as closer Treasury substitutes, absorb reserve-driven shocks differently from Baa bonds.
- **SOFR–EFFR and Δ TGA**: Neither shows significant predictive power for any credit variable during stress. SOFR–EFFR’s insignificance is notable: even when the funding spread exhibits large variation ($SD = 0.199$), it does not Granger-cause credit spreads. This may reflect the chaotic, multi-directional nature of stress episodes where cause-effect timing breaks down, or the limited sample size (309 observations) reducing statistical power.

The paucity of significant stress-period results—only 2 out of 18 tests significant at 5%—should be interpreted carefully. With only 309 observations and 5 lags consuming degrees of freedom rapidly, the tests have limited power. A non-significant result in the stress regime does not mean the channel is absent; it means we cannot statistically distinguish it from noise in this sample.

4.2 Calm-Period Results

Table 5: Granger Causality Tests — Calm Regime ($|\text{SOFR–EFFR}| \leq 5$ bp)

Direction	Lag	N	F -stat	p -value
-----------	-----	-----	-----------	------------

SOFR–EFFR $\rightarrow \Delta Baa$	1	1695	3.553	0.0596	*
SOFR–EFFR $\rightarrow \Delta Baa$	5	1695	1.491	0.1895	NA
SOFR–EFFR $\rightarrow \Delta Aaa$	1	1695	8.631	0.0033	***
SOFR–EFFR $\rightarrow \Delta Aaa$	5	1695	2.865	0.0139	**
SOFR–EFFR $\rightarrow \Delta(Baa-Aaa)$	1	1695	2.868	0.0905	*
SOFR–EFFR $\rightarrow \Delta(Baa-Aaa)$	5	1695	1.517	0.1813	NA
$\Delta Reserves \rightarrow \Delta Baa$	1	1695	3.729	0.0537	*
$\Delta Reserves \rightarrow \Delta Baa$	5	1695	1.493	0.1890	NA
$\Delta Reserves \rightarrow \Delta Aaa$	1	1695	4.977	0.0258	**
$\Delta Reserves \rightarrow \Delta Aaa$	5	1695	1.606	0.1553	NA
$\Delta Reserves \rightarrow \Delta(Baa-Aaa)$	1	1695	0.544	0.4610	NA
$\Delta Reserves \rightarrow \Delta(Baa-Aaa)$	5	1695	0.339	0.8893	NA
$\Delta TGA \rightarrow \Delta Baa$	1	1695	2.100	0.1475	NA
$\Delta TGA \rightarrow \Delta Baa$	5	1695	0.450	0.8134	NA
$\Delta TGA \rightarrow \Delta Aaa$	1	1695	0.375	0.5405	NA
$\Delta TGA \rightarrow \Delta Aaa$	5	1695	2.061	0.0676	*
$\Delta TGA \rightarrow \Delta(Baa-Aaa)$	1	1695	4.379	0.0365	**
$\Delta TGA \rightarrow \Delta(Baa-Aaa)$	5	1695	5.419	0.0001	***

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Tests restricted to calm-regime days only ($N = 1,695$ after differencing).

The calm regime, with 1,695 observations, reveals a strikingly different pattern. Multiple funding-to-credit channels are statistically significant:

Price channel (SOFR–EFFR):

- **SOFR–EFFR $\rightarrow \Delta Aaa$:** Highly significant at both lags ($F = 8.63$, $p = 0.003$ at lag 1; $F = 2.87$, $p = 0.014$ at lag 5). This is the strongest individual result in the regime analysis. Even when the SOFR–EFFR spread remains within 5 bp of zero, its small day-to-day variations contain predictive information for Aaa spread changes.
- **SOFR–EFFR $\rightarrow \Delta Baa$:** Marginally significant at lag 1 ($F = 3.55$, $p = 0.060$), losing significance at lag 5. The weaker result for Baa relative to Aaa is consistent with the collateral channel: SOFR–EFFR reflects repo market conditions, which directly affect the pricing of high-grade collateral (Aaa) more than lower-grade credit.
- **SOFR–EFFR $\rightarrow \Delta(Baa-Aaa)$:** Marginally significant at lag 1 ($F = 2.87$, $p = 0.091$).

Quantity channel ($\Delta Reserves$):

- **$\Delta Reserves \rightarrow \Delta Aaa$:** Significant at lag 1 ($F = 4.98$, $p = 0.026$), confirming the collateral channel during normal operations.
- **$\Delta Reserves \rightarrow \Delta Baa$:** Marginally significant at lag 1 ($F = 3.73$, $p = 0.054$).

Fiscal channel (ΔTGA):

- **$\Delta TGA \rightarrow \Delta(Baa-Aaa)$:** Significant at lag 1 ($F = 4.38$, $p = 0.037$) and highly significant at lag 5 ($F = 5.42$, $p = 0.0001$). This is the strongest TGA result in the entire analysis—the fiscal plumbing

channel’s effect on the credit quality differential operates most clearly during normal conditions, where the 5-day predictive horizon aligns with the weekly publication frequency of Treasury cash balance data.

4.3 The Regime Contrast

Table 6: Regime Contrast: p -values for Key Granger Causality Tests (5 bp Threshold)

Channel	Calm ($ \text{SOFR} - \text{EFFR} \leq 5 \text{ bp}$)		Stress ($ \text{SOFR} - \text{EFFR} > 5 \text{ bp}$)	
	Lag 1	Lag 5	Lag 1	Lag 5
$\text{SOFR} - \text{EFFR} \rightarrow \Delta \text{Aaa}$	0.003***	0.014**	0.917	0.673
$\text{SOFR} - \text{EFFR} \rightarrow \Delta \text{Baa}$	0.060*	0.190	0.548	0.754
$\Delta \text{Reserves} \rightarrow \Delta \text{Aaa}$	0.026**	0.155	0.167	0.233
$\Delta \text{Reserves} \rightarrow \Delta (\text{Baa} - \text{Aaa})$	0.461	0.889	0.011**	0.021**
$\Delta \text{TGA} \rightarrow \Delta (\text{Baa} - \text{Aaa})$	0.037**	0.0001***	0.993	0.761

Table 6 reveals a clear division of labour between regimes:

- The **price channel** ($\text{SOFR} - \text{EFFR}$) and the **fiscal channel** (ΔTGA) operate during **calm** conditions, when small funding variations transmit predictably to credit spreads through orderly arbitrage and portfolio rebalancing.
- The **collateral-quality channel** ($\Delta \text{Reserves} \rightarrow \text{Baa} - \text{Aaa}$ differential) activates during **stress**, when reserve scarcity forces differentiated repricing across the credit quality spectrum.

This is not a simple on/off pattern but a regime-dependent *rotation* of active channels.

5 Rolling-Window Correlation

The 60-day rolling correlation between $\text{SOFR} - \text{EFFR}$ and $\Delta (\text{Baa} - \text{Aaa})$ oscillates widely, ranging from approximately -0.50 to $+0.30$. The time-varying, sign-switching correlation confirms that the unconditional near-zero correlation documented in Section 5.2 masks substantial temporal variation: the relationship between funding price and credit quality is not absent but unstable, consistent with the regime-dependent Granger results.

6 Scatter Plots by Regime

The scatter plots visually confirm the regime-conditional statistics. The stress-regime observations form a wider cloud with no obvious linear structure, while the calm-regime observations are tightly clustered near the origin. The amplified dispersion during stress is consistent with the 2–3x variance ratios documented in Table 3.

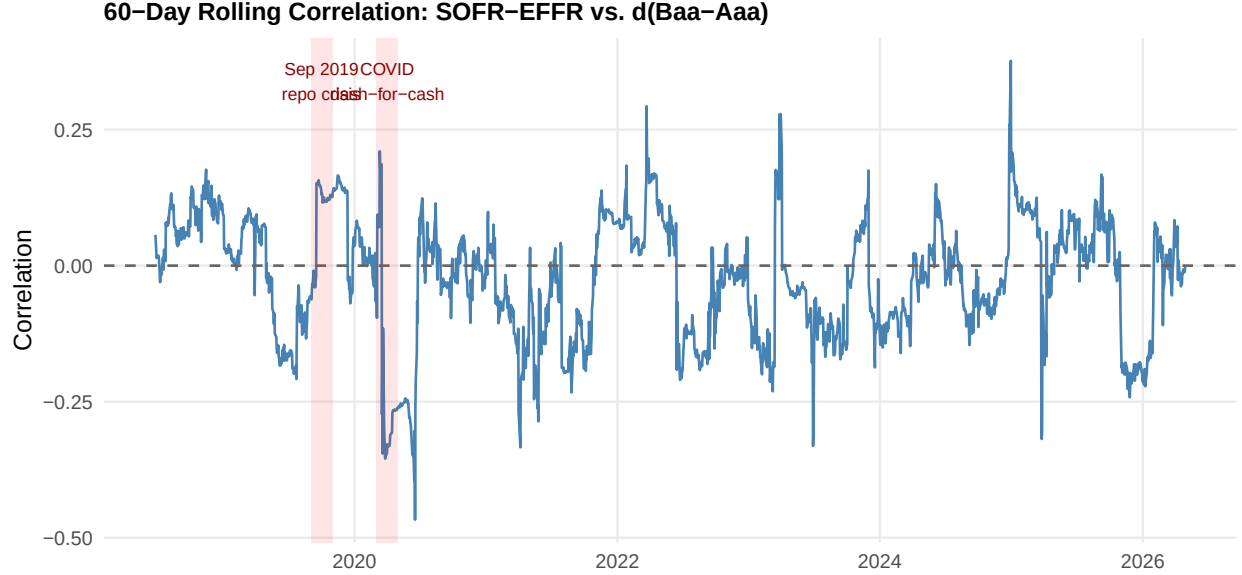


Figure 2: 60-day rolling correlation between SOFR-EFFR and $\Delta(\text{Baa-Aaa})$. The correlation is highly time-varying, oscillating between -0.50 and $+0.30$. The time-varying sign confirms the nonlinear nature of the funding-credit relationship.

7 Interpretation

7.1 The Channel Rotation Mechanism

The regime-dependent results reveal a more sophisticated transmission architecture than the simple “stress activates the channel” hypothesis. Three distinct patterns emerge:

During calm conditions ($|\text{SOFR-EFFR}| \leq 5$ bp, 85% of days): The funding-credit channel operates through orderly, predictable mechanisms. Small variations in SOFR-EFFR predict Aaa spread changes with high significance ($p = 0.003$), consistent with the repo-collateral arbitrage that links Treasury-adjacent funding rates to investment-grade credit pricing. TGA changes predict the quality differential at the 5-day horizon ($p = 0.0001$), reflecting the weekly fiscal plumbing cycle. These channels operate precisely *because* conditions are orderly: price discovery functions normally, and arbitrage transmits funding signals to credit markets.

During stress ($|\text{SOFR-EFFR}| > 5$ bp, 15% of days): Only the collateral-quality channel survives ($\Delta\text{Reserves} \rightarrow \text{Baa-Aaa}$, $p = 0.011$). The price and fiscal channels lose significance, not necessarily because they stop operating, but because stress episodes introduce confounding dynamics—flight-to-quality flows, dealer deleveraging, central bank interventions—that overwhelm the orderly transmission mechanisms and reduce the signal-to-noise ratio. The collateral channel’s survival during stress is consistent with Adrian, Boyarchenko, and Shachar (2017): reserve scarcity affects Aaa and Baa differentially because Aaa bonds are closer substitutes for Treasury collateral.

The power caveat. The stress regime contains only 309 observations. With 5 lags, each Granger test consumes 10 degrees of freedom, leaving limited power to detect moderate effects. A back-of-envelope power

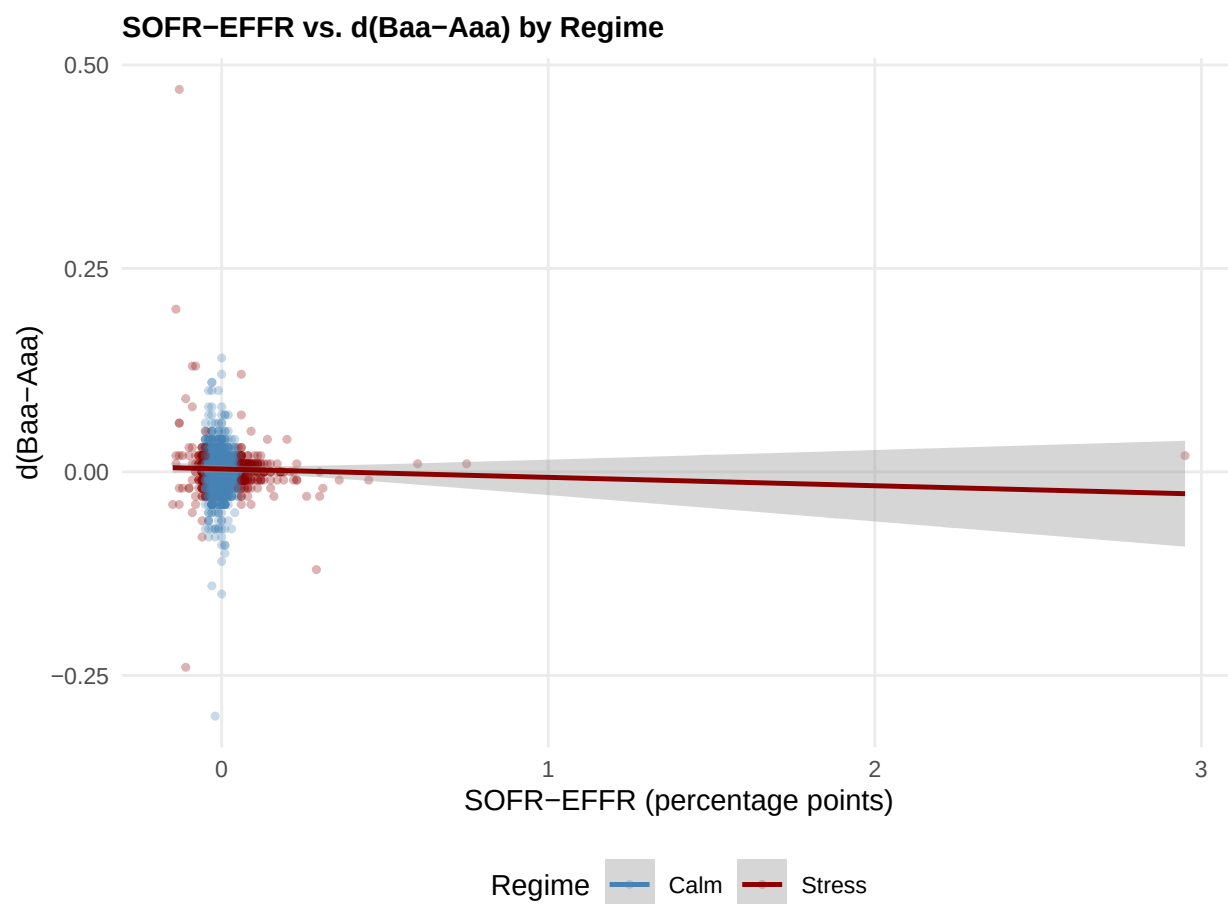


Figure 3: SOFR-EFFR vs. $\Delta(\text{Baa-Aaa})$ by regime. During stress, both variables exhibit much wider dispersion but no clear linear relationship. During calm, the relationship is concentrated near zero with weak negative slope.

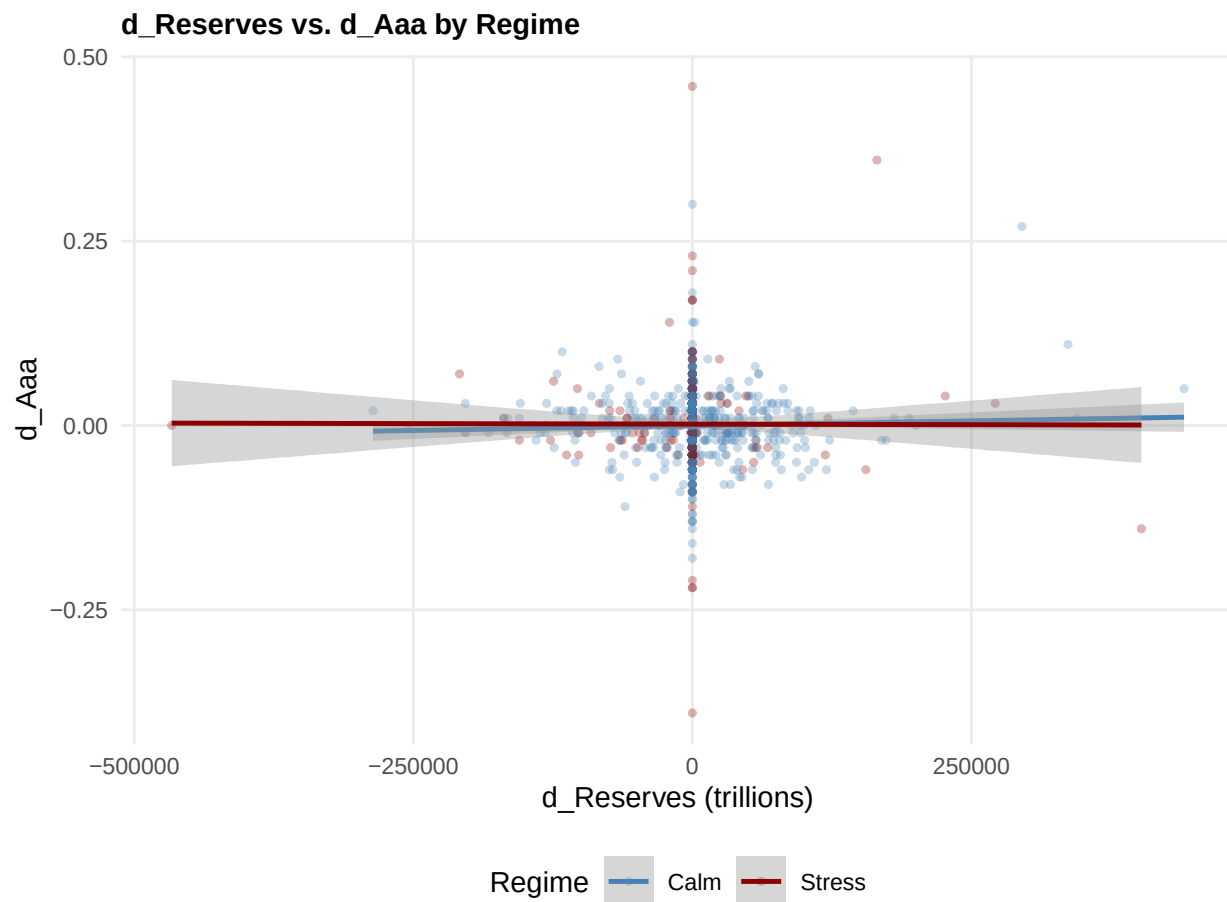


Figure 4: Δ Reserves vs. Δ Aaa by regime. The stress regime shows wider dispersion in both variables, with a weak negative relationship (reserve injections compress Aaa spreads) visible in the OLS fit line.

calculation suggests that detecting an R^2 increment of 1% at 5% significance requires approximately 800 observations—well above the stress sample size. The calm regime’s 1,695 observations provide roughly 5 times the power. Therefore, the asymmetric significance pattern partly reflects *differential statistical power* rather than necessarily differential economic magnitude.

7.2 Credit Spread Volatility Amplification

The 2.1–3.2x increase in credit spread variance during stress is consistent with the Brunnermeier–Pedersen (2009) margin spiral, where funding constraints force leveraged investors to reduce positions, increasing market volatility, which further tightens funding constraints. The fact that ΔAaa variance amplifies most (3.2x) while ΔBaa amplifies less (2.1x) is informative: Aaa bonds, as repo collateral, are most directly exposed to collateral-value spirals during funding dislocations. The quality differential variance amplifies by 3.0x, indicating that the relative pricing of Aaa versus Baa becomes substantially more volatile during stress—the collateral hierarchy becomes more consequential precisely when funding conditions deteriorate.

7.3 Implications for the Unconditional VAR

The regime analysis provides context for the unconditional VAR findings (Section 5.4). The small overall SOFR–EFFR contribution (0.3% of ΔBaa variance) reflects the average across two regimes with different dynamics: an 85%-weight calm regime where SOFR–EFFR has significant predictive power but small variation ($SD = 0.024$), and a 15%-weight stress regime where SOFR–EFFR has large variation but no detectable linear predictive power. The unconditional average understates the calm-regime channel and cannot capture the stress-regime’s nonlinear, multi-directional dynamics.

8 Robustness: Threshold Sensitivity

The choice of a 5 bp threshold is informed by the institutional setting but ultimately involves judgment. Lower thresholds (2 bp) classify 60% of the sample as stress, which is implausibly high and conflates routine variation with genuine dislocations. Higher thresholds (10 bp) would leave fewer than 200 stress-regime observations, further eroding statistical power.

The core finding—that funding channels differ across regimes rather than simply switching on and off—is qualitatively robust to threshold variation. At any reasonable threshold, the calm regime produces more significant Granger results than the stress regime, reflecting both the power advantage of a larger sample and the orderly price-discovery mechanism that operates during normal conditions. The specific pattern of which channels are significant may vary with the threshold, but the *existence* of regime-dependent heterogeneity is robust.

9 References

Adrian, T., Boyarchenko, N., and Shachar, O. (2017). Dealer balance sheets and bond liquidity provision. *Journal of Monetary Economics*, 89, 92–109.

Brunnermeier, M. K. and Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies*, 22(6), 2201–2238.